

Platform-Enabled People Governance 2.0

The PEPG Board: An Open-Source Problem Infrastructure for Distributed Public Governance

Author: Milad Habibpour

Email: milad.habibpour

Version: 2.0 — Revised Conceptual Model

Abstract

Digital technologies create new possibilities for citizen participation, collaborative problem-solving, civic monitoring, and public-sector innovation. Yet participation is often fragmented across institutions, social networks, consultation systems, and digital platforms, while public problems lack a common, persistent, and auditable representation. This paper presents a revised version of Platform-Enabled People Governance (PEPG) through the introduction of the PEPG Board, an open-source and potentially global problem infrastructure. The Board provides common governance services including problem registration, verification, account integrity, problem-definition revision, voting, provenance, relationship mapping, and auditability. Solutions, discussions, implementation, and other participation may occur on external platforms or institutional systems while remaining linked to a common Problem ID. A central contribution is the treatment of problem definition as a revisable governance object: participants may propose sub-problems, improved definitions, separate problems, or challenges to an insufficiently supported definition. PEPG 2.0 combines common problem infrastructure with distributed participation while retaining legitimate institutional authority where legal responsibility, public resources, verification, and accountability require it.

Keywords: Platform-Enabled People Governance; PEPG; digital governance; civic technology; participatory governance; problem-centered governance; problem definition; citizen participation; open source; GovTech; public accountability.

1. Introduction

Contemporary public administration increasingly faces problems that cross institutional, geographic, professional, and technological boundaries. Infrastructure deterioration, environmental degradation, transportation failures, inefficient services, and regulatory problems cannot always be addressed effectively through isolated administrative departments.

Citizens, experts, businesses, civil-society organizations, researchers, and local communities possess substantial information and problem-solving capacity outside formal government structures. Digital technologies can connect these distributed capabilities to public decision-making. The OECD identifies multiple forms of citizen participation across the policy cycle and includes issue identification and problem definition among early stages of participation.

Digital participation also creates fragmentation. A single public problem may be discussed simultaneously on social networks, messaging applications, government portals, local websites, academic platforms, and informal community groups. People may have many places to discuss a problem but no common place where the problem itself is persistently defined, verified, revised, related to other problems, and publicly tracked.

PEPG 2.0 addresses this gap by introducing the PEPG Board while preserving PEPG as a governance model rather than reducing it to a single software platform.

2. From PEPG 1.0 to PEPG 2.0

The initial PEPG concept proposed a people-centered and digitally enabled governance model connecting citizens, government institutions, experts, funding, implementation, and oversight around public problems. Further development revealed several challenges.

Challenge 1 — Complete institutional independence. Public governance cannot realistically assume complete independence from legitimate public institutions. Government bodies retain responsibilities involving legal authority, public budgets, regulation, security, statutory decisions, and public accountability. PEPG therefore does not seek to replace the state; it seeks to connect state capacity with distributed social capacity.

Challenge 2 — Platform fragmentation. If PEPG is only a model, a public problem may exist simultaneously across many disconnected platforms without a stable identity. The PEPG Board provides a common problem infrastructure.

Challenge 3 — Fixed problem definitions. The first formulation of a problem may be incomplete, imprecise, or based on insufficient information. PEPG 2.0 therefore treats problem definition as provisional and revisable.

Challenge 4 — Trust and identity. Open participation creates risks including duplicate accounts, impersonation, coordinated manipulation, unverifiable claims, and artificial voting. PEPG 2.0 therefore introduces account-integrity and verification mechanisms.

3. The Revised PEPG Concept

PEPG 2.0 is defined as a problem-centered governance model that enables distributed participation while providing a common open infrastructure for the registration, verification, evolution, and coordination of public problems.

Two components should be distinguished. The PEPG Model is the governance logic that can be adopted by communities, institutions, cities, regions, or countries. The PEPG Board is an open-source infrastructure that provides common services for public problems. PEPG is therefore not synonymous with the PEPG Board; the Board is an implementation infrastructure for the model.

4. The PEPG Board

The PEPG Board is a public problem registry and governance layer. Its basic function is to give each problem a persistent identity. For example, a problem may receive the identifier PEPG-0002 together with the title “Poor Asphalt Quality on an Abstract Street in Tanzania.” The Problem ID becomes the reference point for subsequent activities.

A problem record can contain its definition, verification status, related problems, revision proposals, votes, external discussions, solutions, implementation references, oversight information, and reports. This creates a persistent object around which distributed activity can be coordinated.

5. The Board Is Not the Whole Participation Environment

A core principle of PEPG 2.0 is the separation of governance services from participation spaces. A person may discover a problem on X, discuss it on Telegram, collect evidence through a local community, develop a technical solution with engineers, and implement a project through a government or private system. PEPG does not need to replace these environments.

External activities can be linked through a common Problem ID. The Board therefore becomes a coordination and trust layer rather than an attempt to host every activity itself. This also reduces technological lock-in.

6. Core Principles of the PEPG Board

Problem-Centeredness. Every PEPG process begins with a public problem, which becomes the primary organizing object.

Persistent Problem Identity. Every registered problem receives a unique identifier that remains associated with it even when its description evolves.

Open Registration. People should be able to propose problems without prior institutional membership, while submission remains distinct from verification.

Transparency and Provenance. Important changes should preserve their origin, time, rationale, and relevant decision process.

Contestability. Participants should be able to challenge problem definitions, verification states, and other significant classifications through defined procedures.

Separation of Functions. Problem identification, verification, funding, implementation, and oversight should not unnecessarily be controlled by one actor.

7. Problem Verification

A problem can have multiple verification states: Unverified → Community Verified → Authoritatively Verified.

An unverified problem has been submitted but has not received sufficient confirmation. A community-verified problem has received sufficient independent evidence or confirmation under predefined rules. An authoritatively verified problem has been confirmed by a recognized authority such as a municipality, provincial administration, ministry, university, regulatory body, professional organization, or other recognized institution.

Verification of a problem should remain separate from endorsement of a solution. An authority may verify that a road is damaged without determining which repair method should be adopted.

8. Account Integrity

PEPG can distinguish levels of account integrity, such as unverified account, verified account, and verified organization. Risk states such as under review, suspicious, or restricted may also be used under transparent rules.

The purpose is not necessarily to force every participant to publicly disclose their identity. It is to reduce artificial multiplication of influence through duplicate, automated, or coordinated accounts. PEPG should therefore distinguish identity privacy from participation integrity.

9. Governance of Problem Definition

A central innovation of PEPG 2.0 is that a problem definition is treated as a provisional governance object rather than an immutable statement created by the first proposer.

For example, a citizen may submit “Poor asphalt quality on Street X.” Another participant may argue that the underlying problem is premature pavement deterioration caused by inadequate drainage and non-compliant construction. The second participant can propose a revised definition without silently overwriting the original.

The proposal becomes a Problem Definition Revision subject to transparent review. This creates a governance process around the framing of public problems themselves.

10. Four Types of Problem Definition Proposals

Sub-Problem. The proposal represents a smaller problem contained within the original problem.

Improved Definition. The proposal describes the same problem more accurately or precisely.

Separate Problem. The proposal represents a distinct issue and should receive a new Problem ID while preserving its relationship to the original.

Incorrectly Defined Problem. The participant argues that the original problem is insufficiently supported, misleading, or incorrectly formulated. This creates a review process rather than silently deleting the original.

11. Approval and Revision Procedures

If the original proposer accepts a revision, the proposed definition can become the accepted definition through a transparent proposer-approval path. If the proposer rejects it, the proposal can enter community review.

If it reaches a predefined legitimacy threshold under the Board's governance rules, the revised definition may become authoritative. The exact threshold should be established through further research and experimentation rather than arbitrarily fixed in the initial model. This balances proposer rights with the principle that no individual should obtain permanent control over the definition of a public problem.

12. Definition History

PEPG should not simply erase previous definitions. Significant changes should be preserved as part of the problem's history: v1 — Original Definition; v2 — Proposed Revision; v3 — Community-Approved Revision; v4 — Authority-Verified Definition.

This creates semantic provenance: users can identify who originally defined the problem, who proposed a change, why it was changed, who supported the change, when the change occurred, and whether the result was community- or authority-verified.

13. Voting

Voting can be used for specific governance functions, including problem-definition revisions. However, PEPG should not reduce all participation to simple popularity scores.

Participants should be able to indicate whether a proposal is a sub-problem, improved definition, separate problem, evidence that the original problem is incorrectly defined, or insufficiently supported. Account-integrity mechanisms should reduce artificial manipulation.

14. External Participation and Solutions

PEPG deliberately allows solutions and other activities to occur elsewhere. Solution development may happen in a messaging group, professional network, Git repository, university environment, private organization, or public institution. The Board can record the relevant external reference while maintaining the common Problem ID.

This preserves PEPG as a model that communities can adopt without forcing every community to use the same communication platform. PEPG services provide the trust, identity, voting, verification, provenance, and linkage functions needed to coordinate distributed participation.

15. Trust and Verification Services

The PEPG Board can provide common services that external platforms may not provide consistently: factual verification, account integrity, structured voting, provenance, problem relationships, revision history, and audit trails.

These services should be modular. A PEPG implementation may use different technical approaches for identity, voting, verification, and storage while conforming to the same governance principles.

16. Optional Blockchain Integrity Layer

Blockchain can be used where appropriate as an integrity layer rather than as a mandatory foundation of PEPG. Potential records include problem-definition revisions, verification events, contracts, reports, legislative inputs, voting records, and timestamps.

The governing principle is technological neutrality: PEPG should not depend on a particular technology such as blockchain, artificial intelligence, or a particular social network. Technology should serve the governance requirement rather than define the model.

17. Relationship Between Government and PEPG

PEPG does not assume that government institutions disappear. Public authorities can participate as accountable actors and, where appropriate, trusted verification authorities. A provincial government may verify a problem; a university may verify technical evidence; citizens may propose solutions; a private company may implement a contract; and independent participants may monitor implementation.

PEPG is therefore not anti-government. It is better understood as a framework for reducing unnecessary concentration of governance capacity while retaining legally defined institutional responsibilities.

18. Preventing Concentration of Power

No single actor should unnecessarily control the complete chain of problem definition, verification, solution selection, funding, implementation, oversight, and reporting. Separating these functions can create institutional and informational checks against concentration and conflicts of interest.

This principle also applies to the digital infrastructure itself. The Board's source code, rules, moderation procedures, and decision mechanisms should be open to inspection and accountable governance.

19. Speed and Large-Scale Coordination

A problem-centered architecture may reduce unnecessary sequential processing by allowing multiple actors to work around a shared problem object simultaneously. Experts, funders, citizens, implementers, and oversight actors can potentially engage in parallel rather than waiting for every stage to finish before the next begins.

The potential benefit is therefore not merely faster communication but the parallelization of governance capacity. This may be valuable for large infrastructure projects, emergency responses, complex public-service problems, and strategic initiatives. These benefits should nevertheless be treated as hypotheses for empirical testing rather than guaranteed outcomes.

20. Open Source as Institutional Infrastructure

The PEPG Board is well suited to open-source development. Public source code and openly documented rules can support inspection, independent auditing, community contributions, reduced vendor lock-in, reproducibility, and local adaptation.

GitHub can serve as an initial development environment for a prototype, including source code, governance rules, documentation, issue discussions, research, and proposed changes. The development process can therefore embody some of the openness principles that PEPG promotes.

21. PEPG as a Model Rather Than a Single Product

A city could implement PEPG through its own government infrastructure. A university could use it to organize research problems. A community could operate an open-source Problem Board. A national government could deploy an institutional instance. An international organization could create a global registry.

The underlying principles remain consistent while institutional and technological implementations vary. PEPG is therefore a governance model that can be implemented through different institutional and

technical arrangements.

22. Governance Principles of the PEPG Board

The revised model proposes the following principles: openness; persistent problem identity; provisional and revisable definitions; provenance; verification separation; identity integrity; distributed participation; institutional accountability; separation of functions; contestability; historical continuity; and technological neutrality.

Together, these principles are intended to make the problem lifecycle more transparent without requiring all participation to occur in a single application.

23. Theoretical Contribution

The primary theoretical contribution of PEPG 2.0 is the combination of problem-centered governance, distributed participation, and governable problem definition.

Many governance frameworks ask who decides. PEPG additionally asks: Who defines what the problem actually is? This matters because problem framing can influence which solutions are considered possible. By making problem definition transparent, contestable, revisable, and historically traceable, PEPG creates a governance layer around the formation of the problem itself.

24. Relationship to Existing Participatory Governance

PEPG should not be interpreted as a replacement for established participation mechanisms. Rather, it can provide infrastructure connecting them. The OECD identifies multiple participation methods across the policy cycle, including issue identification, policy formulation, decision-making, implementation, and evaluation.

PEPG extends this logic by giving the problem itself a persistent digital identity that can remain associated with those stages and with activities occurring across different platforms.

25. Risks and Limitations

PEPG 2.0 does not eliminate the challenges of digital governance. Important risks include misinformation, coordinated manipulation, unequal digital access, dominance by highly active users, institutional capture, majority tyranny, privacy violations, identity surveillance, algorithmic bias, false verification, political pressure, platform dependency, and cybersecurity threats.

An open-source platform is not automatically democratic. Its algorithms, verification rules, voting mechanisms, moderation procedures, identity policies, and governance authority must themselves be transparent, reviewable, and contestable.

26. Research Agenda

Future research should investigate: (1) optimal voting mechanisms; (2) detection of coordinated manipulation; (3) privacy-preserving identity verification; (4) verification thresholds; (5) governance of trusted authorities; (6) dispute-resolution procedures for problem definitions; (7) safeguards against majority domination; (8) digital inclusion; (9) cross-platform interoperability; (10) blockchain and non-blockchain provenance mechanisms; (11) empirical measurement of decision-making speed; and (12) comparative implementation across institutional contexts.

PEPG 2.0 should initially be treated as a research and prototyping program rather than a fully validated governance solution. Open-source experimentation can provide a practical basis for testing its assumptions.

27. Conclusion

The first formulation of PEPG emphasized the ability of digital networks to connect citizens, institutions, expertise, funding, implementation, and oversight around public problems. PEPG 2.0 recognizes that this model requires a stronger foundation.

The revised model introduces the PEPG Board as a common, open-source and potentially global infrastructure in which public problems can receive persistent identities, verification states, relationship structures, revision histories, voting mechanisms, account-integrity indicators, and links to external participation spaces.

At the same time, PEPG does not require all solutions, discussions, implementation activities, or oversight to occur inside the Board. This distinction allows PEPG to remain both a governance model and an extensible infrastructure.

The most significant conceptual contribution is the treatment of problem definition as a governed and revisable object. A public problem is not necessarily permanently defined by the person who first reports it. It can be challenged, refined, decomposed, related to other problems, or replaced through transparent procedures involving the proposer, the community, and, where appropriate, trusted authorities.

Ultimately, PEPG does not propose that government should disappear, that technology should replace institutions, or that every form of participation should occur on one platform. It proposes that public governance can be organized around persistent, transparent, contestable problems while the people, institutions, expertise, resources, and technologies needed to address those problems remain distributed.

PEPG 2.0 is therefore problem-centered, people-enabled, state-compatible, platform-flexible, open-source, and continuously revisable.

References

- OECD. (2022). OECD Guidelines for Citizen Participation Processes. OECD Publishing. <https://doi.org/10.1787/f765caf6-en>.
- OECD. (2021). Eight Ways to Institutionalise Deliberative Democracy. OECD Publishing. <https://doi.org/10.1787/4fcf1da5-en>.
- World Bank. (2025). GovTech Maturity Index 2025 Update. World Bank.
- World Bank. Global Program on GovTech & Public Sector Innovation. World Bank.
- United Nations. (2024). United Nations E-Government Survey 2024. UN DESA.

Author Note

This paper presents a conceptual and research-oriented formulation of PEPG 2.0. The proposed mechanisms are intended for further theoretical development, open-source prototyping, and empirical evaluation rather than as claims of already-established causal effectiveness.